

Demonstration of Petting Interaction with Breathing Cat Using Mid-Air Ultrasound Haptic Feedback

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ABSTRACT

This study introduces a petting interaction system with a virtual breathing cat, utilizing mid-air ultrasound haptic feedback to enhance the realism of pet experiences in virtual reality (VR). Previous research has focused mainly on the tactile presentation of the fur texture, and interaction with realistic virtual animals has not been fully investigated. Therefore, based on the prior work, the proposed system employs airborne ultrasound tactile displays (AUTDs) and a hand tracking camera to deliver real-time, non-contact haptic feedback simulating not only the soft fur texture but also the breathing motion. By dynamically controlling the ultrasound focal point's position and intensity, the system reproduces the tactile sensations of fur stroking and rhythmic respiratory motion. The visual component features a 3D cat model synchronized with breathing cycles, situated in a familiar domestic setting. User experience design focused on creating a calming interaction, with tactile feedback adjusting based on hand movement patterns.¹

Index Terms: affective haptics, animal-assisted therapy, fur display, mid-air haptics, virtual reality

1 INTRODUCTION

Interacting with animals such as cats and dogs is comforting and enjoyable, and has been utilized in the fields of animal-assisted therapy and entertainment [7, 2]. In particular, simulated pet experiences, such as robotic pets and pet games in virtual reality (VR), have been extensively studied due to their advantages, including reduced caregiving burdens, ethical responsibilities, and allergy concerns. Researchers have focused on enhancing these experiences by making them more realistic and lifelike. For instance, in interactions with pet-type robots, it has been shown that simulating the breathing motion of living creatures through visual and haptic feedback can enhance relaxation effects [5].

Although pet-type robots can effectively present realistic fur textures and breathing motions, they are limited by hardware constraints regarding the types of fur and animal species, and they also face challenges in achieving the dynamic motion and flexibility seen in real animals. In contrast, VR-based pet interactions offer high flexibility in terms of the variety of pets and their behaviors. However, conventional haptic feedback approaches, which typically use vibrotactile actuators, struggle to realistically convey the soft fur texture of real animals. To overcome these limitations, recent research has introduced mid-air ultrasound haptic feedback, enabling non-contact tactile sensations that more closely mimic the experience of stroking real fur [3]. Building on this prior work, this study proposes an advanced VR pet interaction system by simulating soft fur texture and incorporating dynamic breathing mo-

tions. This combined approach aims to achieve a highly realistic and calming pet interaction experience.

2 METHODS

2.1 Apparatus

In this study, a head-mounted display (Meta Quest 3, Meta) was utilized for delivering both visual and auditory stimuli in the virtual environment. A hand tracking camera (Leap Motion Controller 2, Ultraleap) was integrated to continuously capture real-time hand data, including hand movements and shape. For delivering tactile stimuli, two Airborne Ultrasound Tactile Displays (AUTDs) [4, 6] were interconnected, each comprising an array of 249 ultrasonic transducers per unit (Figure 1).

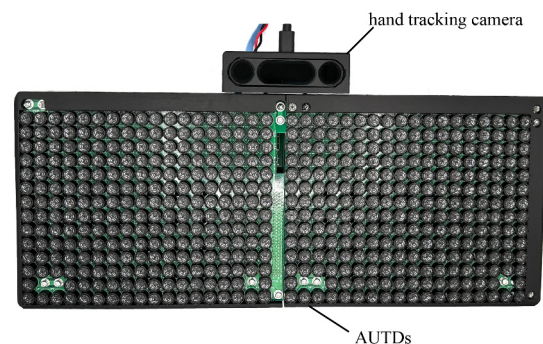


Figure 1: Top view of the system with hand tracking camera and two airborne ultrasound tactile displays

The integration of a hand tracking camera with the AUTD system facilitates the precise, real-time generation of ultrasonic focal points on the user's palm, thereby enabling the delivery of customizable tactile feedback.

By modulating the position and intensity of these focal points, various textures and shapes can be represented. For instance, in a modulation method known as Spatio-Temporal Modulation (STM) [1], the perceived roughness can be altered by changing the speed at which the focal point moves in a circular pattern. Using this approach, previous research successfully presented the tactile sensations of stroking fur [3]. In this study, we similarly control the position and intensity of the focal point to achieve the interaction with a breathing cat.

2.2 Design of Haptic Feedback

In tactile interactions with animals, such as placing one's hand on a breathing cat, the perception of respiratory movement is primarily influenced by the rhythmic expansion and contraction of the animal's body. This periodic movement modulates the contact area and intensity of tactile feedback perceived by the hand. Specifically, during the inhalation phase, the animal's body inflates, resulting in an increased contact area and enhanced tactile pressure against the palm. Conversely, during exhalation, the contraction of the body reduces both the contact area and pressure.

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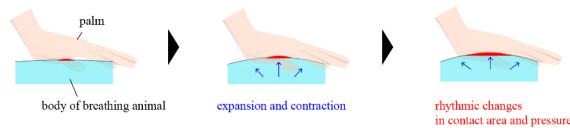
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¹<https://youtu.be/oQrccIH301Y>

In prior studies on simulating respiratory motion in pet-type robots, the up-and-down motion of the surface was mechanically reproduced [5]. This study proposes presenting the rhythmic changes in the contact area and pressure of respiratory motion by controlling the diameter of circular motion of the focal point and amplitude of ultrasound haptic stimulation (Figure 2). Specifically, during inhalation, both the diameter of the circular motion and the amplitude are increased, whereas during exhalation, both the diameter and amplitude are decreased. The pattern of changes was set to a sawtooth wave, following prior research. In the user experience described in the next section, the radius on the palm was adjusted within a range of approximately 0.5 cm to 2 cm, and the amplitude was ranged from approximately 4 mN/cm² to 15 mN/cm² (Figure 3).

haptic perception of respiratory movement



design of mid-air ultrasound haptic feedback

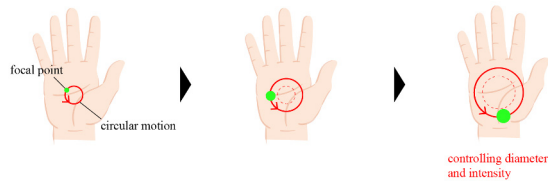


Figure 2: Haptic perception of respiratory motion and design of mid-air ultrasound haptic feedback

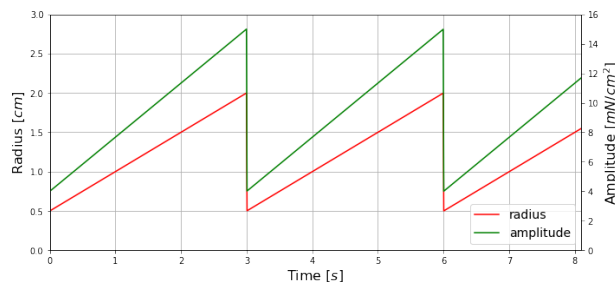


Figure 3: Radius and Amplitude of mid-air ultrasound haptic stimulation in user experience

In previous research on presenting the tactile sensation of fur, differences in tactile sensation between stroking along and against the natural direction of growth were achieved by adjusting the speed and intensity of focal movement [3]. This study aims to extend that approach by incorporating simulated breathing motion. To achieve this, thresholds were set based on the speed of hand movement. When the hand moves, the system provides the tactile sensation of stroking fur, while when the hand remains still on the animal, it switches to simulating the breathing motion under the palm.

3 USER EXPERIENCE

For the user experience, we designed a relaxing interaction with a breathing cat. For the visual representation, we prepared a 3D model of a cat whose skin and fur dynamically expand and contract in synchronization with its breathing cycle. The virtual environment depicted a familiar domestic scene—a cat resting on a bed in

a typical household setting. The sleeping cat exhibited slow breathing motions at a rate of approximately 20 cycles per minute, synchronized with snoring sounds. Furthermore, the system allowed users to interact naturally by moving their hands. Using data from the hand tracking camera, the user's hand movements were continuously captured, and a corresponding hand model was rendered in real time (Figure 4).

The system enables users to freely touch the virtual cat. Haptic stimuli, including the tactile sensations of fur texture and rhythmic movements simulating breathing, are rendered in real time to the user's palm based on the interaction type. This setup is expected to deliver a highly realistic and pleasant interaction with the virtual cat. Future enhancements could focus on increasing the interaction's flexibility, potentially broadening its application to various user groups and enhancing relaxation effects.



Figure 4: VR scene when user's palm is placed on the back of a cat

ACKNOWLEDGMENTS

This work was supported by JSPS KAKENHI Grant Numbers JP23H04333, JP21H03478

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